

Circuit Layout Problems

1. Identify what the main aims are of good circuit layout and discuss how stray inductance can be minimised for circuits that carry a significant amount of current (such as power electronic converters).
2. Using a boost converter circuit as an example, show how you could use capacitor based global ac decoupling to reduce the stray inductance area associated with the high di/dt current loop.
3. What benefits are provided through the use of Ground and Power planes, and how do we manage these planes on an individual PCB to minimise the effect high frequency and high power functional groups have on sensitive electronics?